

ECSE 4320/6320: Advanced Computer Systems

Electrical, Computer, and Systems Engineering
Rensselaer Polytechnic Institute

Course Information

Course Title: Advanced Computer Systems
Course Number: ECSE 4320/ECSE 6320
Semester and Year: Fall 2025
Credit Hours: 3
Meeting Times: Mondays and Thursdays 2:00~3:20PM
Classroom: JEC 6309

Instructor

Prof. Tong Zhang
Office: CII 6217
Email: tzhang@ecse.rpi.edu
Office Hours: Mondays 1~2PM

TA: Garrett Gagnon
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Office Hours: Wednesdays 9:30~10:30AM, CII 6118

Course Description

Developing a broad and solid understanding of modern computer systems is essential not only for designers who design and implement computing infrastructure (e.g., data center, enterprise appliance, and mobile devices), but also for programmers and developers. This course will cover advanced topics and current industrial trends of modern computer systems from both hardware and software aspects. This course will first present key design principles and trends of modern computing hardware platform, including instruction/data/thread-level parallelism in CPU and GPU, cache and memory hierarchy, DRAM and non-volatile memory technologies, and heterogeneous and disaggregated computing architecture. This course will then discuss software design techniques and methodologies tailored for modern computing hardware platform, including consistency, concurrency control, transaction, indexing and key-value store, and storage model. This course will also discuss the future trends in the semiconductor and computing industry in the post-Moore era.

Pre-requisites

ECSE 2660 CANOS or CSCI 2500 Computer Organization, or equivalent with permission from the instructor

Reading Materials

- John Hennessy and David Patterson, “Computer Architecture: A Quantitative Approach”, 2017
- Bruce Jacob, David Wang, and Spencer Ng, “Memory Systems: Cache, DRAM, Disk”, 2007
- Martin Kleppmann, “Designing Data-Intensive Applications”, 2017
- A list of research papers

Student Learning Outcomes

1. Understand the core design concepts and principles of computer systems, and the trends in the computing industry
2. A deeper understanding of advanced computer system hardware stack, including CPU architecture, cache and memory, and data storage

3. A deeper understanding of the key software system design concepts, including consistency, concurrency, transaction, and indexing
4. Understand the state of the art in data management systems, and distributed computing systems
5. Strengthen the ability to carry out independent research, and gain experience on making technical presentations

Additional Student Learning Outcomes for 6320

1. Develop an ability to carry out in-depth analysis of research papers and identify potential improvements
2. Develop an ability to write up the research problem and analysis results as a short paper or technical report

Course Assessment Measures

Course assessment will be done through in-class and on-line participation and discussion, and a series of individual projects.

Grading policy

	4000-level	6000-level
1. In-class participation and discussion	10%	10%
2. Course design projects	90%	80%
3. Research report*		10%

* Students taking the course at the 6000 level are required to submit a research report or a technical report, which is not required of the 4000-level students (listed as 2 under ***Additional Student Learning Outcomes for 6000 Level*** above).

Academic Integrity

Student-teacher relationships are based on trust. For example, students must trust that teachers have made appropriate decisions about the structure and content of the courses they teach, and teachers must trust that the assignments that students turn in are their own. Acts which violate this trust undermine the educational process. Please refer to the *Rensselaer Handbook of Student Rights and Responsibilities* and *The Rensselaer Graduate Student Supplement* for definitions of various forms of academic dishonesty and the applicable penalties.

In this class, all the in-class presentations must represent the student's own work. In cases where help was received, or teamwork was allowed, a notation should indicate your collaboration. Submission of any assignments that is in violation of this policy will result in a *penalty of removing the entire grade of the assignment*. Repeat violations will be reported to the office of the Dean of Students. Teamwork may be allowed the final project, but only after discussion with the instructor.